Supplementary Material – Physiological Signal Processing

Beyond Productivity: Measuring Developers' Cognitive Load During GenAI-Supported Software Development

I. Physiological Signal Processing and Pipeline Selection

This supplementary material provides the detailed physiological signal pre-processing and pipeline selection procedures of the main paper. To eliminate confounding artefacts from the data, the physiological signals were pre-processed before analyses. The processing steps were performed per participant and working day and are described in the following.

A. Acceleration (ACC)

The wrist acceleration data were used to filter motion-related artefacts in the physiological signals. The data of the three movement axes x, y and z were independently filtered with a 4th-order Butterworth bandpass filter with a low cutoff at 0.5 Hz and a high cutoff at 4 Hz. Subsequently, the root mean square (RMS) was calculated across the three axes as a measure of total movement magnitude (1).

$$\mathrm{RMS} = \sqrt{\frac{(x^2 + y^2 + z^2)}{3}} \quad (1)$$

Two motion threshold strategies were evaluated on the BVP data for identifying and removing motion artefacts. The best performing strategy was then also applied to the other physiological signals. The first strategy used a hard cutoff defined as the sum of the mean and standard deviation of the RMS values (2). Physiological data coinciding with RMS values exceeding the cutoff were classified as motion artefacts. The second strategy employed a percentile-based cutoff with the 90th percentile of the participant-specific RMS distribution for each working day used as the threshold. Consequently, physiological data coinciding with RMS values above the threshold were classified as motion artefacts.

Hard cutoff:

$$T_{hard} = \mathrm{mean}(\mathrm{RMS}) + \mathrm{std}(\mathrm{RMS}) \quad (2)$$

90th percentile cutoff:

$$T_{90} = P_{90}(\mathrm{RMS}) \quad (3)$$

B. Blood Volume Pulse (BVP)

We compared six BVP pre-processing pipelines that differed in their extent of signal pre-processing (TABLE .): The *Baseline* pipeline solely performs peak detection on the raw BVP signal, from which the HR and HRV metrics are calculated.

The *Neurokit* pipeline applies standard pre-processing steps as implemented by the widely used NeuroKit2 library [1], including signal cleaning (nk.clean), peak detection on the cleaned signal (nk.ppg_findpeaks), and correction of erroneous peak detections based on outliers in the peak-to-peak differences (nk.signal_fixpeaks). Neither pipeline explicitly addresses the problem of motion artefacts in the data.

The *Motion* and *Motion_90* pipelines extend the *Neurokit* pipeline by incorporating motion artefact removal based on wrist acceleration. High-motion segments are identified by using participant-specific movement thresholds (T_{hard} or T_{90}). These segments are eliminated and reconstructed using monotone cubic interpolation. Then the *Neurokit* processing steps are applied. The pipelines differ only in their motion-thresholding strategy.

The *HR_Pipeline* and the *HR_Pipeline_90* further extend the previous pipelines by additionally filtering implausible heart rate transitions. Peaks associated with successive HR changes >40% are removed. These pipelines also differ only in their motion thresholding strategy.

After the pre-processing steps, HR and HRV metrics were calculated for each working task. For all pipelines, HR and HRV metrics were only computed for working task segments that contained >20% peaks which were associated with physiologically valid HR ranges (40-180 bpm).

The following metrics were calculated per working task and are used in the following analyses:

- **mean_HR**: mean HR (bpm)
- **median_HR**: median HR (bpm)
- **HRV_SDNN**: standard deviation of inter-beat intervals (IBIs), reflecting overall heart rate variability (both sympathetic and parasympathetic influences)
- **HRV_RMSSD**: root mean square of successive differences between adjacent IBIs, reflecting parasympathetic activity
- **HRV_pNN50**: percentage of adjacent IBI pairs that differ more than 50ms, reflecting short-term HRV and parasympathetic regulation

TABLE I. BVP Pre-Processing Pipeline Comparison

Pipeline	Motion filter	Type of motion filter	Abrupt changes in HR filtered	>20% Valid HR peaks (40-180 bpm)	BVP signal processing steps
Baseline	✗	-	✗	✓	Peak detection on raw data: nk.ppg_findpeaks(raw_data)
Neurokit	✗	-	✗	✓	Neurokit's pipeline: nk.clean, nk.ppg_findpeaks, nk.signal_fixpeaks
Motion	✓	T_{hard}	✗	✓	Neurokit's pipeline + motion artefact filtering and signal interpolation
Motion_90	✓	T_{90}	✗	✓	Neurokit's pipeline + motion artefact filtering and signal interpolation
HR_Pipeline	✓	T_{hard}	✓	✓	Motion artefact removal pipeline + filtering of consecutive HR beats that differ >40%
HR_Pipeline_90	✓	T_{90}	✓	✓	Motion artefact removal pipeline + filtering of consecutive HR beats that differ >40%

C. Pre-Processing Pipeline Evaluation Based on BVP Data

To obtain reliable results, the impact of data quality and pre-processing steps of the physiological data on the analyses was examined. There is no consensus in the research community on which exact processing steps should be performed, but processing steps affect data quality and therefore may influence the results derived from the physiological data. The extent of preprocessing required depends strongly on the recording devices and tasks performed with the device – especially when body movement is involved.

For pre-processing pipeline selection, the six pipelines were compared by fitting the same linear mixed-effects model to each pipeline (4). The model included the log-transformed and participant-standardized RMSSD (lnRMSSD_z) physiological metric, GenAI use (yes/no), task category (development-heavy, collaboration-heavy and other), and task duration (in minutes) as fixed effects, with participant as a random intercept. The random intercept was included to account for repeated working task observations within participants and between-participant differences in cognitive load ratings. To improve compliance with the assumptions of the linear mixed-effects models, physiological metrics with skewed distributions (e.g., RMSSD) were natural log-transformed and z-standardized per participant prior to analysis.

$$\text{Cognitive load} \sim \text{lnRMSSD_z} + \text{GenAI} + \text{task category} + \text{task duration} + (1|\text{participant}) \quad (4)$$

To examine the impact of data quality and pre-processing on the results, the pre-processing pipelines were compared using identical linear mixed-effects models of cognitive load based on lnRMSSD_z, task category, GenAI use and task duration. The model comparisons were conducted with lnRMSSD_z as an exemplary physiological HRV metric. Other metrics showed comparable results. Only working tasks, for which all pipelines produced valid HR and HRV metrics, were compared. The common subset contained 89 working tasks.

Model performance was highly comparable across pipelines with only negligible differences in model fit (AIC: 302.73-303.12) and explanatory power (marginal R^2: 0.316-0.320; conditional R^2: 0.374-0.375) (TABLE II.). This indicates that the choice of the suggested pre-processing pipeline on our physiological BVP data will not have a major impact on the cognitive load predictions and result interpretation in the following.

The pipelines, however, differed substantially in the number of retained working tasks (TABLE III.). The number of retained working tasks was calculated separately for each pipeline based on the complete set of 382 working tasks. A working task was counted as retained when valid HR and HRV metrics could be calculated for the respective pre-processing pipeline. The mean HR, RMSSD and SDNN values in TABLE III. were calculated across the working tasks retained by the respective pipeline and across participants.

The *Motion* pipeline retained the largest proportion of working tasks (68.85%; 263 out of 382), whilst also maintaining comparable model fit and mean HR and HRV values (mean HR=70.16; mean RMSSD=0.28; mean SDNN=0.21) to the other pipelines (TABLE III.).

The substantially lower number of retained tasks for *HR_Pipeline* (45.03%; 172 tasks) and *HR_Pipeline_90* (23.82%; 91 tasks) and the lower mean RMSSD (HR_Pipeline=0.14, HR_Pipeline_90=0.13) and mean SDNN (HR_Pipeline=0.15, HR_Pipeline_90=0.14) values suggest that the more aggressive filtering may have removed physiological variability in addition to artefacts.

On the other hand, the reduced number of retained tasks when running the *Baseline* pipeline (45.55%, 174 tasks), together with higher mean RMSSD (0.33) and mean SDNN (0.23) values, indicate insufficient artefact removal.

In sum, the *Motion* pipeline with motion artefact removal via a hard cutoff value provided the best trade-off between data retention, data quality improvement due to movement artefact removal, physiological plausibility, and processing extent and complexity. The *Motion* pipeline was therefore selected for BVP pre-processing. The corresponding T_{hard} motion threshold was also used for motion artefact removal in the EDA and TEMP signals described in the following sections.

II. Processing of Additional Physiological Signals

A. Electrodermal Activity (EDA)

Consistent with the motion artefact removal strategy from the BVP signal, the same T_{hard} cutoff was also applied to the EDA signal. Signal segments exceeding the threshold were removed, and the resulting gaps filled via monotone cubic interpolation. The cleaned EDA signal was then decomposed into its tonic and phasic components using the NeuroKit2 [1] decomposition method. Skin conductance responses (SCRs), visible as peaks in the phasic component of the EDA signal, were detected when their amplitude surpassed 0.05 μS. Following pre-processing, EDA metrics were then calculated per working task from the cleaned EDA signal, as well as from the tonic and the phasic components to capture both slow-varying and event-related EDA:

- **mean_SCL**: mean skin conductance level (SCL), i.e. the mean of the tonic signal component
- **mean_phasic**: mean of the phasic signal component, reflecting the average magnitude of skin conductance fluctuations
- **mean_cleaned_EDA**: mean of the cleaned EDA signal containing both the tonic and phasic signal components
- **SCRs_norm_duration**: number of SCRs normalized by the task duration in seconds, representing the rate of phasic electrodermal responses

The EDA metrics reflect the sympathetic nervous system activity, with high values indicating increased physiological arousal. In the context of working task performance, elevated values have been associated with increased mental workload and stress-related states.

B. Skin Temperature (TEMP)

Motion artefacts were also removed from the TEMP signal by using the T_{hard} cutoff. TEMP data below 25°C and above 40°C were removed, since the values are physiologically implausible. The gaps in the data were filled via monotone

cubic interpolation. TEMP metrics were then calculated from the pre-processed signal per working task:

- **mean_TEMP**: mean skin temperature
- **std_TEMP**: standard deviation of the skin temperature
- **range_TEMP**: range of skin temperature values

TABLE II. COMPARISON OF PRE-PROCESSING PIPELINES BASED ON LINEAR MIXED-EFFECTS MODEL PERFORMANCE ON A COMMON DATASET (MODEL FIT: AKAIKE INFORMATION CRITERION (AIC); EXPLANATORY POWER VIA MARGINAL R^2 AND CONDITIONAL R^2)

Pipeline comparison lnRMSSD	AIC	Marginal R^2	Conditional R^2
Baseline	303.07	0.316	0.375
Neurokit	302.73	0.320	0.374
Motion	302.74	0.320	0.374
Motion_90	302.77	0.320	0.375
HR_Pipeline	303.07	0.317	0.374
HR_Pipeline_90	303.12	0.316	0.374

TABLE III. PROPORTION AND NUMBER OF RETAINED WORKING TASKS (OUT OF 382 TASKS) AND MEAN HR, RMSSD AND SDNN VALUES ACROSS ALL WORKING TASKS AND PARTICIPANTS PER PRE-PROCESSING PIPELINE

Pipeline	Retained tasks	Mean HR (bpm)	Mean RMSSD	Mean SDNN
Baseline	45.55% (174)	71.59	0.33	0.23
Neurokit	65.45% (250)	70.56	0.28	0.21
Motion	**68.85% (263)**	**70.16**	**0.28**	**0.21**
Motion_90	47.64% (182)	71.24	0.31	0.22
HR_Pipeline	45.03% (172)	72.04	0.14	0.15
HR_Pipeline_90	23.82% (91)	73.29	0.13	0.14

REFERENCES

[1] Dominique Makowski, Tam Pham, Zen J. Lau, Jan C. Brammer, François Lespinasse, Hung Pham, Christopher Schölzel, and S. H. Annabel Chen. 2021. NeuroKit2: A Python toolbox for neurophysiological signal processing. *Behav. Res. Methods* 53, 4 (August 2021), 1689–1696. https://doi.org/10.3758/s13428-020-01516-y